Branch Specialization in Attention

Phase 1 Repeat-Match Causal Ablation Checkpoint

Research checkpoint

May 21, 2026

Current Question

Question: Do structurally corresponding attention heads have stable causal functions across random seeds?

The current concrete target is the early-layer repeat-match role found in Pythia-160M. Previous results showed that this role is concentrated and much more stable after raw-score head alignment than by same index.

This checkpoint asks whether those aligned heads are causally important for a repeated-token prediction task.

Ablation Design

Evaluation sequences:

$$[x_1, \ldots, x_{32}, x_1, \ldots, x_{32}]$$

Loss is scored on second-half continuation positions $x_i \to x_{i+1}$. The same 64 synthetic sequences are reused across all nine seeds.

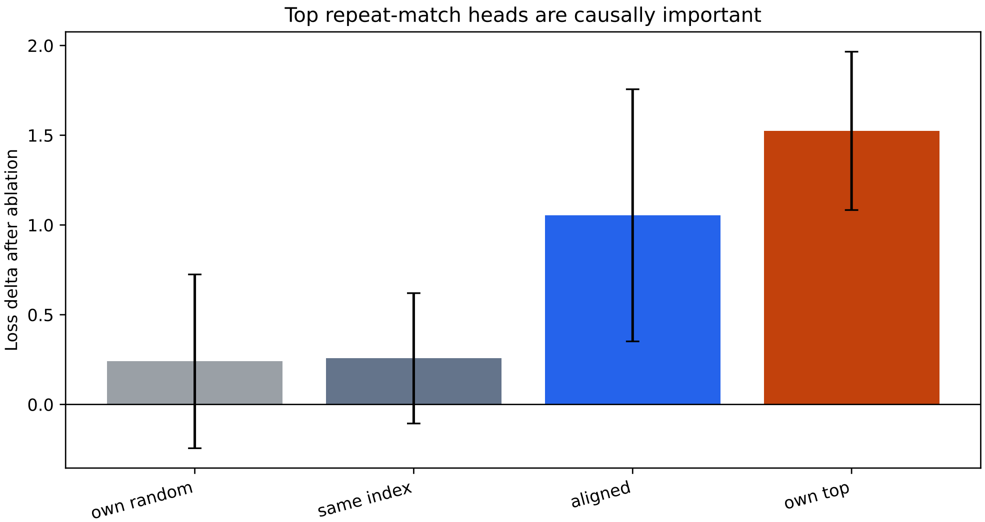
Ablation zeros selected per-head attention outputs before the GPT-NeoX attention output projection.

Conditions

- **own top**: target seed's top repeat-match heads in layers 0–1.
- **own random**: random same-layer controls, excluding top heads.
- **same index**: another seed's top head indices ablated in the target.
- **aligned**: another seed's top heads mapped into the target by raw-score Hungarian alignment.

Positive loss delta means the ablation made repeated-token prediction worse.

Aggregate Result

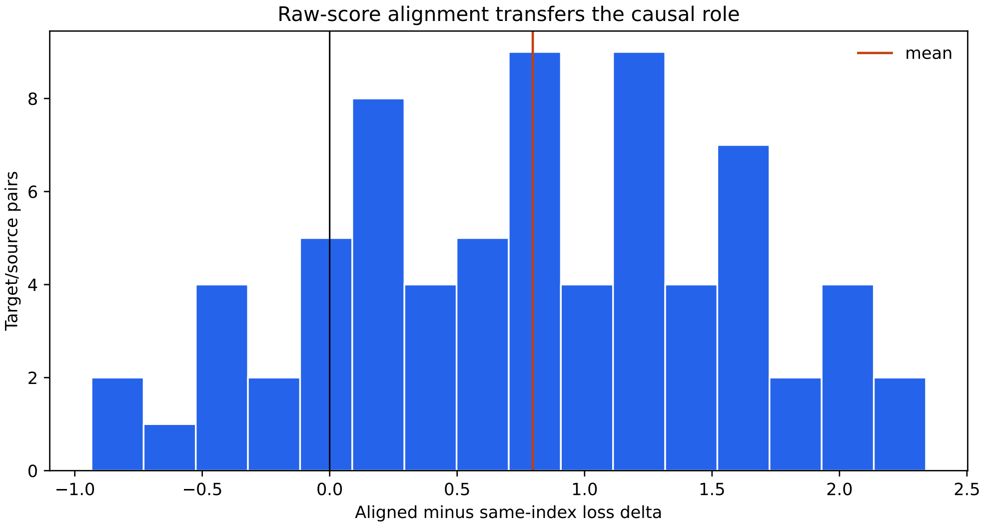


Numerical Summary

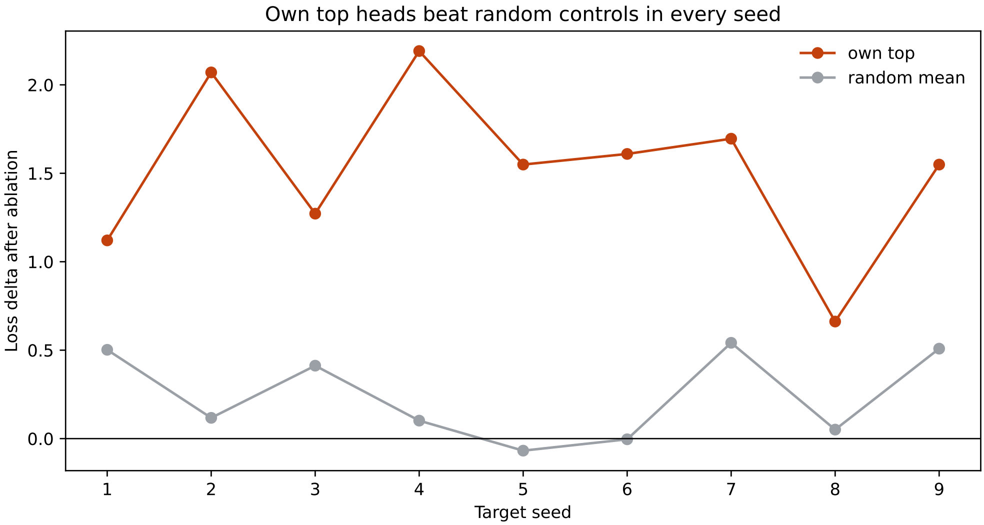
Condition	Mean loss delta	Mean target-logit delta
own random	0.2403	-0.7184
same index	0.2571	-0.7014
aligned	1.0538	-1.5750
own top	1.5244	-2.0689

Aligned transferred heads are far closer to own-top heads than to same-index or random controls.

Transfer Test



Own-Seed Causal Check



Interpretation

Findings:

- Repeat-match heads are causally relevant on this synthetic task.
- Same layer/head index is a poor cross-seed transfer rule.
- Raw-score alignment transfers the causal role much better.

Stronger claim not yet established:

- This does not yet test heterogeneous branch architectures.

Decision

Continue with repeat-match / induction-style behavior as the first causal test bed.

Next step:

- build or adapt a small branched-attention training setup;
- vary branch structure or structural heterogeneity;
- measure raw-score alignment, role specialization, and causal ablation with the same pipeline.

Provenance

- Script: `scripts/repeat_match_ablation.py`
- Output: `results/phase1_repeat_match_ablation_pythia160m`
- Report: `doc/phase1_repeat_match_ablation_pythia160m.md`
- Role input: `results/phase1_pythia160m_attention_role_specialization`
- Alignment input: `results/phase0_pythia160m_seed1_to_9_step143000_raw_scores`
- Data copied into this deck folder under `data/`